

R Task 1

Objective

Implement the Gaussian Mixture Model (GMM) using the Expectation-Maximization (EM) algorithm from scratch in Base R.

Step 1

Generate a synthetic dataset with two Gaussian clusters (100000 observations each). Save it as `synthetic_data.csv`, display the first few observations, and plot the dataset.

Step 2

Initialize $K = 2$, mixing coefficients, mean vectors, and covariance matrices.

Step 3

Create a custom Multivariate Gaussian Probability Density Function.

Step 4

Perform the Expectation Step (E-Step) by calculating the responsibility matrix.

Step 5

Perform the Maximization Step (M-Step) by updating weights, means, and covariance matrices.

Step 6

Repeat the EM algorithm until convergence or a maximum of 20 iterations while storing log-likelihood values.

Step 7

Generate and save `dataset.png`, `final_clusters.png`, and `log_likelihood.png`.

Project Files

Main R Program: **task 1 using r.R**

Dataset: `synthetic_data.csv`

Output Folder: `dataset.png`, `final_clusters.png`, `log_likelihood.png`

README.md

R task1.pdf

Expected Output

Display the final weights, means, covariance matrices, cluster assignments, and the generated plots.